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ColdFlux Superconducting EDA and TCAD Tools Project: Overview and Progress

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Abstract— The IARPA SuperTools program requires the development of superconducting electronic design automation (S-EDA) and superconducting technology computer aided design (S-TCAD) tools aimed at enabling the reliable design of complex superconducting digital circuits with millions of Josephson junctions. Within the SuperTools program, the ColdFlux project addresses S-EDA and S-TCAD tool research and development in four areas: (i) RTL synthesis, architectures and verification; (ii) analog design and layout synthesis; (iii) physical design and test; and (iv) device and process modeling/simulation and cell library design. Capabilities include, but are not limited to: device level modeling and simulation of Josephson junctions, modeling and simulation of the superconducting process manufacturing processes, powerful new electrical circuit simulation, parameterized schematic and layout libraries, optimization, compact SPICE-like model extraction, timing analysis, behavioral, register-transfer-level and logic syntheses, clock tree synthesis, placement and routing, layout-versus-schematic extraction, functional verification, and the evaluation of designs in the presence of magnetic fields and trapped flux. ColdFlux consists of six research groups from four continents. Here we present an overview of the current and planned activities related to the project, and justify design assumptions and decisions that were made to allow the development of design tools for million-gate circuits.

Index Terms—Design automation, Flux pinning, Integrated circuit synthesis, Process modeling, Superconducting integrated circuits

I. INTRODUCTION

THE ColdFlux project, which falls under the IARPA SuperTools program [1], is focused on the development both front-end and back-end superconducting electronic design automation (S-EDA) and technology computer-aided design (TCAD). The eventual goal of the project is to enable very large

scale integration (VLSI) design and verification of Superconductive Electronics (SCE) as a step toward the development of energy-efficient, scalable high performance computers. As a proof-of-concept demonstration, the ColdFlux team will undertake the design of a 64-bit Reduced Instruction Set Computer (RISC) microprocessor with the tools and cell libraries developed under the project. For this project, we limited the supported dc-biased logic families to Rapid Single Flux Quantum (RSFQ) [2] for proof-of-concept and ERSFQ [3] for energy efficiency. Support for ac-biased logic families is limited to Adiabatic Quantum Flux Parametron (AQFP) [4].

ColdFlux results are predominantly open-source, so that many of our design decisions are influenced by the availability of open-source modules.

In this paper we present the scope and intended deliverables of the ColdFlux project, list advantages above current state-of-the-art and report on the progress made thus far.

II. RTL SYNTHESIS, ARCHITECTURES AND VERIFICATION

A. High-level and Register-Transfer Level (RTL) synthesis

Logic synthesis is the process of transforming a set of Boolean equations into a network of gates. We use ABC [5] as our logic synthesis platform and add SFQ-specific synthesis modules.

We have developed various logic optimization techniques, done design of multi-input RSFQ cells, and performed retiming to minimize the overhead of path balancing D Flip-Flops (DFFs) in SFQ designs [6]. We have developed two SFQ specific technology-independent logic optimization techniques [7]:

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balanced_rewriting (brw) and balanced_refactoring (brf). Both techniques are effective in reducing the total gate count. We have also developed two algorithms for performing technology mapping of SFQ circuits with the goal of minimizing the DFF path balancing overhead and logical depth [8]. The logic optimization procedures are incorporated into our synthesis tool called qPALACE, which will be released to the public soon.

B. RTL simulation and verification

We use the open-source Verilog simulator Icarus Verilog, or iVerilog [9], to verify synthesized circuits.

C. Design tool integration and IDE

The ColdFlux tool chain is designed as a collection of individual modules that work at least under Linux CentOS 7. Schematic and layout editing are done with XIC [10], and simulation, analysis, optimization and layout extraction modules are integrated with XIC. Tool design is modular to enable development and maintenance between around 30 developers over four continents. Most modules run from the terminal, with command line parameters and configuration files used to pass information and command and control functionality.

D. Datapath synthesis and architectural optimization

One demonstrative end goal of the project is to use the S-EDA tools to synthesize a RISC-V processor built from superconducting cell libraries. To this end we will explore potential (micro-)architecture enhancements to account for the limitations and opportunities of the new technology. From an architecture view point, the fundamental challenge with SFQ circuits is gate level pipelining, which can cause even simple arithmetic logic unit (ALU) operations such as multiply to take several cycles to complete. While the throughput of such a system can be maximized to operate at one operation per cycle through careful design, in the targeted CPU single threaded performance requires data dependent operations to be separated at least by the ALU latency. Since the hardware cannot be burdened with this task, we have been working on purely compiler-based methods to create independent instruction sequences. Rather than rely on programming languages such as C++, we believe a new domain-specific language (DSL) is necessary to allow programmers to express instruction independence, similar to how CUDA allowed programmers to specify thread independence. While there is no current effort in DSL design, such an effort will have significant positive impact on performance. For now, we plan to hand optimize code placement for several kernels to demonstrate the performance on the RISC-V processor.

Reducing the producer-consumer delay in an ALU pipeline can also be accomplished by allowing eager data forwarding at the bit granularity. In this approach an ALU may forward results of its partial computations to the consumer instruction in a pipelined manner. Rather than waiting for the worst-case latency for computing the full result, a consumer instruction may start its execution earlier by operating on partial results that are forwarded from the producer. For instance, the least n-

significant bits of computation from a producer can be used by the consumer. Such an approach requires ALU designs that forward partial results. To this end, we are exploring bit-skewed ALU design computation on the data bit can be staggered to enable eager forwarding.

We also explore the memory limitation of superconducting devices as an architecture challenge. Since designing new memory architectures is outside of the scope of this project, we plan to use existing methodologies to design register files as the primary on-chip storage. Two types of register file can be implemented: a non-destructive read out (NDRO) register file and a much more area efficient destructive read out (DRO) register file. A DRO register file, which maximizes on-chip storage, loses its value after every read. However, most programs use registers to store frequently accessed data that may be read more than once. To account for these two competing demands, we are designing a heterogeneous register file consisting of a small set of NDRO registers and a larger number of DRO registers. We are exposing the register file heterogeneity to our compiler framework, which is under development and built on LLVM, to maximize the usage efficiency. From preliminary analysis we believe that a compiler can isolate single use registers from multi-use registers (registers that are read more than once) and thus can exploit the heterogeneous register file design.

E. Verification framework

Our open-source verification framework consists of formal, semi-formal and simulation-based tools, and algorithms, with partial reliance on the existing verification methodologies and open-source tools. There are however several key differences between SFQ and CMOS logic circuits, related to pulse signaling, gate fanout, ultra-deep (gate-level) pipelining, and path balancing, which should be considered. Our tool for logical equivalence checking (LEC) [11] first performs structural verification of the DUV (design under verification). In case of a successful pass on both path balancing and fanout count, functional checking is performed using conventional LEC. If the path balancing check fails, further processing is required, as a circuit may not be fully balanced, yet function correctly. The DUV is therefore transformed to its accurate clocked-based model. This transformation will unify structural and functional information of the SFQ circuit. A clocking control logic unit is also added to control the inputs and a logical miter circuit is then constructed. Next a Boolean satisfiability (SAT) [12] instance is added to encode the miter circuit and finally a SAT solver is used to analyze the miter. To improve the verification scalability and coverage, we also plan to release a semi-formal verification platform based on UVM (Universal Verification Methodology) that can accelerate coverage directed test generation by running assertions in gate level and RTL abstractions [13]. We will also release a model checking [14] tool, which is capable of formally proving complex temporal properties at RTL that will mainly target control-intensive circuits. To validate our framework, we have also developed several SFQ circuit benchmarks that include faulty designs with various structural and functional errors

III. ANALOG DESIGN AND LAYOUT SYNTHESIS

A. JoSIM Simulation engine

Although JSIM [15] and WRSpice [10] are available as open-source simulation engines for superconducting circuits, we developed a new superconducting circuit simulation engine, JoSIM [16], for ColdFlux. JoSIM is the first simulation engine to support both voltage and phase-based simulation, so that it supports zero-voltage current flow in superconducting loops and can process our compact models where static magnetic fields couple to inductive loops with zero resistance [17].

In addition to the standard Resistively and Capacitively Shunted Junction (RCSJ) model [18] [19], we aim to support the Werthamer microscopic tunnelling junction model [20] of the Josephson junction for more accurate simulation of AQFP [4] circuits with unshunted junctions and RSFQ circuits at ultimate clock frequencies. JoSIM thus combines the capabilities of JSIM and PSCAN [21] [22].

Other features of JoSIM include direct support for parameterized elements, optimization, margin analysis and simulation of circuits with more than 100,000 components.

JoSIM v2.0 (October 2018) is already available for use [23].

B. Layout synthesis

Layout synthesis methods are developed to define circuit layouts as scripts with the aid of Python-based parameterized cells (PCells), or in combination with a layout synthesizer, Absynth, that we are developing. Absynth will integrate directly with parameter extraction tools such as InductEx [24] to allow parameter verification [25].

With layouts scripted as parameterized cells, resynthesis is possible when row dimensions are altered, track pitch changed, or process feature reductions become available. We will also integrate parameterized layout scripting and synthesis methods with Layout-versus-Schematic (LVS) and Design-Rule Checking (DRC) methods.

Part of our layout synthesis procedure is the synthesis of Passive Transmission Line (PTL) interconnects as striplines ground plane stitching and through-ground vias along the routing paths generated by the placement and routing tools.

C. Process Design Kit (PDK)

Process design kits consist of a set of files that contain descriptions of the basic building blocks of the process. Our PDK for the MITLL SFQ5ee process [26] already contains:

- 1) Primitive device library: symbols; device parameters.
- 2) Verification decks: InductEx layer definition files calibrated against measured experimental structures [27].
- 3) Technology data: Layer information, display attributes, process constraints.
- 4) Rule files: LEF format of abstracted layout data.
- 5) Simulation models of devices: Josephson junctions.

IV. PHYSICAL DESIGN AND TEST

A. Place-and-route and clock tree synthesis and routing

Our placement tool utilizes a row-based methodology [28]. The placement tool for SFQ logic cells (qPlace) implements global placement using one of the state-of-the-art algorithms, namely SimPL. In each iteration of this algorithm, two placement solutions are generated. 1) A placement solution calculated by solving a quadratic objective function (total HPLW) which produces a lower-bound solution (in terms of total HPWL). 2) A placement solution generated by legalizing lower-bound solution (upper-bound solution in terms of total HPWL) [29]. To discourage overlap in lower-bound solution, cell locations in the upper-bound solution work as anchors, providing pulling forces towards cells in the lower-bound solution. Therefore, in each iteration upper-bound solution of previous iteration changes the quadratic formulation. We implemented this algorithm in C++, improving the overall placement solution and runtime significantly with respect to the implementation of the seedling project. Although this algorithm provides a high-quality global solution, it does not necessarily give high quality solution in terms of longest path length (critical wire length).

Once the placement is done, clock tree synthesis must be performed to propagate the clock signal to all the clocked elements. In SFQ technology, most cells receive a clock signal. Additionally, all the cells have a fan-out of 1, and to propagate the signals to more than one fanout, splitters should be used. Thus, a complete H-tree with n clock sink nodes needs $n-1$ clock splitters [30]. Large number of splitters significantly increase the total area dedicated to the clock network.

Our tool qGDR integrates a global router and a detailed router to fulfill the full-chip routing task. The global router assigns loose connection paths for nets while attempting to minimize the number of used vias. The detailed router follows the global routing results to complete the routing task. Development of the detailed router is based on open-source tool (Qrouter) which greedily searches routing paths regardless of any specific physical characteristics. The feedback system is further embedded in qGDR to automatically increase routing tracks until all nets are routed. Following the MIT-LL SFQ5ee process technology, an 8-bit Kogge-Stone adder (KSA) of 638 nets is routed by qGDR in less than 20 seconds.

B. Fault simulation, test pattern generation and built-in-self-test tools

Compared to CMOS, the power and performance benefits of SFQ come from the use of Josephson junctions instead of small feature sizes, hence defects are not a major cause of chip failures and process variations and other non-idealities become the leading causes of chip failure. Due to the nature of quantized pulse-based operation, even a highly-distorted pulse is still interpreted correctly but the timing of the pulse is affected. Hence, with a specific focus on dynamic timing verification and delay testing, we develop delay fault models and characterize cells

under process variations to identify delay fault excitation conditions, sensitization conditions, and error propagation conditions under the effect of process variations. We then propose a completely new timing independent Automatic Test Pattern Generation (ATPG) paradigm to select target delay sub-paths and generate test patterns that guarantee to excite the worst-case delay along a target delay sub-path. The new ATPG tools based on these methods will be released under ColdFlux. We are also developing a timing aware approach to improve the coverage and effectiveness of the set of test patterns generated by our ATPG.

C. Clocking

In addition to the potential use of zero-skew H-trees (see [31]) for clock distribution networks (CDN), we developed an advanced clocking strategy to provide more scalability and more resilience to variability.

We proposed the hierarchical chains of homogeneous cloverleaves clocking, (HC)²LC, a robust and self-adaptive clocking technique for generic and complex pipelines [32]. The proposed clocking methodology is an asynchronous CDN in the form of a hierarchical set of connected asynchronous loops. (HC)²LC nominally oscillates as fast as a synchronous H-tree, but due to its asynchronous nature, exhibits resilience to variations, slowing down when portions of the network are slow [33]. This clocking scheme inherits its robustness from the spatial correlation of various sources of variations [34] and the timing robustness of traditional counter-flow clocking. It trades off reasonable area and power overheads for higher reliability, as quantified by yield [35], and improved scalability.

D. Timing analysis

We extract timing models for dc-biased RSFQ and ERSFQ cells from JoSIM electrical simulations with TimEx [36], of which version 2.03 (August 2018) is available for use [37]. Flux signatures are analyzed for each loop in a circuit under test, and all possible states—each corresponding to a unique flux signature—are found. Critical and delay timings between every pair of inputs are found. We use TimEx both for timing extraction and verification of cell functionality.

For timing verification, we developed `vcd_assert`, of which version 1.0 is available [38], to expand the capability of open-source iVerilog simulations. This tool allows timing assertion verification on the outputs of iVerilog simulations, with timing specifications provided in Standard Delay Format (SDF), so that both open-source and commercial Verilog simulators can be used for timing analysis.

For the extraction of timing models for ac-biased AQFP logic cells, we developed a separate tool called AQFPTX, of which version 1.0 has been delivered. The core extraction approach is similar to what was proposed in [39], but makes use of redefined timing parameters suitable for SDF file generation and provides a more user-friendly extraction configuration. As with TimEx, the simulation engine is JoSIM, although both extraction tools are compatible with JSIM [15] and JSIM_n [40]. Furthermore, it can be used to verify cell-level functionality. Automatic operation margin analysis of excitation current and dc

offset of AQFP cells with respect to timing parameter dependency is also planned for development.

We already delivered a first-order post place-and-route STA tool, SuperSTA [41].

We also developed a timing characterization method that uses look-up tables (LUTs) [42] that are specifically designed for SFQ logic circuits and is similar to the non-linear delay model (NLDM) based LUTs for CMOS circuits. After investigating timing characteristics of SFQ cells, the following LUTs were chosen: Clock-to-Q delay of clocked cells, input-to-output delay of non-clocked cells (JTL, splitter, and merger) and output pulse width of each cell are stored in 2-dimensional (2D) LUTs indexed by (i) the first series inductance (L_{series}) and (ii) the difference (I_Q) between the critical current and the bias current through the first grounded Josephson Junction after the output of the cell. Setup and hold times are stored in 1-dimensional (1D) LUTs based on the incoming pulse width.

E. Thermal analysis

We are developing post-layout power and thermal analysis methods to determine local heating due to static and dynamic power dissipation, with accurate modeling of thermal resistance on-chip in a cryogenic environment [43]. Compact simulation models will include temperature parameters for every element such as inductors, of which the kinetic component is temperature dependent, and for every device such as a Josephson junction. Temperature support, with thermal noise, at component and device level will be supported in JoSIM. Local temperature is used as well to estimate thermal noise in circuits and provide more accurate simulations based on temperature-dependent physical parameters like critical currents and gap voltage [44].

F. Margin calculation and yield analysis

Robustness of an SFQ gate indicates the tolerance of the gate to variations in electrical parameters of its components. Accordingly, a margin is defined as the amount of acceptable variation in an external parameter (e.g. bias current) or internal parameter (e.g. critical current I_C and inductance values) for which the circuit continues to function correctly.

We have layout-extracted Monte Carlo models [45] and optimization methods that utilize Monte Carlo yield analysis [46]. For large circuits, we employ both traditional margin analysis [47] and probabilistic methods [48]. New optimization tools based on these methods as well as more advanced stochastic methods will be released under ColdFlux.

We also analyze operating margins in the presence of external magnetic fields [49] [50], and link this analysis to compact model extraction and experimental measurements [51] [52].

For AQFP logic, the AQFPTX tool used for timing extraction has a planned feature for also performing automatic operating margin analysis. The tool will be able to analyze how changes in the excitation power/clock and dc offset of AQFP affect the logical functionality of cells as well as the timing parameters. The analysis can generate SDF files for various corners of the operation margins so that one can carry out timing analysis for not just the nominal excitation amplitude and dc offset but also for their corner cases.

We have also developed a novel method for accurate margin calculation of (SFQ) logic cells in a superconducting electronic circuit. This method can be utilized as a figure of merit to estimate the robustness of a logic cell without the need for expensive Monte Carlo simulations. This was achieved through efficient state-space exploration of all variability parameters of a RSFQ cell structure. Using this method, distinct parameter dispersion (DPD) based yield of SFQ cells increases by 55% on average [53], compared with state-of-the-art techniques.

V. TECHNOLOGY CAD AND CELL LIBRARY DESIGN

A. Process and device modeling

Josephson junction parameters as a function of manufacturing steps is of key importance in the verification of VLSI circuit yield. We use Level Set Methods (LSM) [54], included in the FLOXXS process simulation tool extended to support devices and process steps for SCE Nb/Al-AIO_x/Nb processes.

The output of the process simulation tool will be combined with the local equations that rule the thermal and electrical behavior of materials and devices to predict the properties. To start with a first simplified approach we chose a phenomenological electromagnetic model based on the two fluid model, associated with Maxwell laws to model the niobium material at the Schubnikov state where $H_{c1} < H < H_{c2}$ since Josephson junctions work in this state in presence of a magnetic field. For the thermal model of type II superconductors, we started with a simple linear model proposed by Maki [55]. In this model, the thermal conductance of the system depends on the concentration of quasiparticles and phonons in the superconductor lattice. This dependency can be translated into a dependency on the magnetic field of the superconductor.

B. Parameterized DC/AC-biased cell libraries

Cell libraries are required to demonstrate the capability of tools for the ColdFlux project, and to allow verification of the tool chain. A parameterized DC-biased RSFQ and ERSFQ cell library has been developed. Test circuit layouts are being prepared. Cells are parameterized for adjustment of nominal I_C – the value for a Josephson junction in a standard Josephson Transmission Line – from 125 μ A to 250 μ A, and layouts are parameterized to adjust for cell height and cell width. For row-based place-and-route purposes, all cells have fixed height, and width is variable as an integer multiple of the routing track pitch [28]. We use passive transmission line (PTL) for all gate-to-gate interconnects, with track pitch between 7 and 10 μ m depending on the characteristic impedance of the stripline PTL. Drivers and receivers are embedded into gate layouts to reduce junction count and delay times and optimized to reduce reflections [56]. In order to calculate maximum PTL length and maximum number of vias, we are developing high-frequency models of PTL to include the plasma frequency of superconductors and to account for dispersion and attenuation [57] [58].

An AC-biased AQFP standard cell library has been developed. We first designed four basic AQFP gates: buffer, inverter and constant 0 and constant 1. More complicated gates such as

AND, OR and MAJORITY are designed via the *minimalist design approach* [59] using a bottom-up manner. Although the size of different cells varies, for example a buffer occupies an area of 15 μ m $\times$ 20 μ m, whereas the size of an AND gate is 45 μ m $\times$ 27.5 μ m, the ac/clock I/O position is fixed for row-based placing and routing in meander structure [60]. We use microstrip lines for gate-to-gate interconnects. The width of the microstrip line is 2.4 μ m, with 50 Ω impedance, for ac/clock propagation. Data delivery striplines are 5 μ m wide, with impedance of 7 Ω .

C. Layout parameter extraction

For layout extraction we use the InductEx tool suite [24] [25], with additions [17] [61] [62] to support the ColdFlux requirements. InductEx was developed for self and mutual inductance extraction from superconducting circuit layouts, but we have since added support for flux trapping analysis [17] [62], external magnetic fields, and the calculation of ground plane return currents. The result is that InductEx can now extract compact models for circuits in the presence of an electromagnetic environment for post-layout verification of field operating margins and deterioration of operating margins due to bias and return current effects.

We are also developing methods to improve the efficiency of numerical electromagnetic field solvers for calculating current distributions on superconducting integrated circuit structures. The well-known integral equation-based tool, FastHenry, is specifically considered [63]. It employs a multilevel fast multipole algorithm (MLFMA) solver to reduce required memory and accelerate the solution. This MLFMA solver introduces approximations into the system matrix representation, leading to a certain level of error in the output. An alternative, multilevel adaptive cross approximation solver with singular value decomposition recompression (MLACA-SVD) [64] [65] has been developed within FastHenry [66] [67]. Results show that the new solver requires less memory than FastHenry's MLFMA, for the same solution accuracy. It also offers simple, comprehensive control over matrix approximation errors, such that a given solution may be obtained to any desired accuracy. A novel extension to the MLACA-SVD solver is reported [68], which yields a further 30% reduction in required memory, without loss of accuracy.

D. Design rule checking and layout-vs-schematic tools

Layout-versus-schematic (LVS) tools generate netlists from physical layouts (e.g. in GDSII format) and compare these to user-generated schematics for graph isomorphism. If the graphs are isomorphic, LVS is satisfied. If the designs are not isomorphic, the LVS tool must provide effective feedback to layout designer.

We started from previous work on LVS for superconductor integrated circuits [69], and developed a method that meshes layouts with Gmsh and the openCASCADE library and builds graphs from nodes at triangle centers and edges between touching triangles. The graphs are then simplified until edges represent unique components, such as junctions, inductors and resistors.

E. Flux trapping analysis tools

Magnetic flux trapped in superconducting films [70], [71] interact with surrounding superconducting circuits, which can adversely affect circuit operations. Moats can be placed in the superconducting ground plane near sensitive regions to mitigate the effects of flux trapping [72].

An existing three-dimensional numerical solver, TetraHenry [73], has been modified to evaluate the effects of trapped flux on superconducting circuits. The Gibbs potential is used to calculate the probability of flux trapping and the self and mutual inductance between moats, holes and applied external magnetic fields. The extracted mutual inductances can then be used within SPICE simulations to determine the operating margins and yield of the circuit for different combinations of trapped vortices within moats. With flux trapping analysis tools now available, a tool is under development to optimize the geometry (size and placement of moats and location of dc bias pads) so that yield margins can be improved.

F. Compact models

Physics driven device modeling is typically slow, so that empirical models (or compact models) that do not directly model the underlying physics are more useful for SPICE simulations.

Apart from the inclusion of temperature dependence in component values for compact SPICE models, we also include the electromagnetic environment through the effects of static magnetic fields, coupling from bias and ground return currents and trapped flux in the compact SPICE models. Compact model extraction for the electromagnetic environment is directly supported by InductEx.

VI. CONCLUSION

The ColdFlux project is delivering back-end tools for better design, optimization and layout of SFQ circuits in magnetic environments, with RSFQ, ERSFQ and AQFP libraries for rapid prototyping of circuit designs by new designers. Front-end tools for synthesis, placement and routing, with clock synthesis for both dc-biased and ac-biased logic families are also being delivered. Complemented with verification tools, from cell layout extraction to RTL verification and post place-and-route timing verification, the ColdFlux tool suite will be aimed at enabling faster, higher yield development and fabrication of SFQ VLSI systems.

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